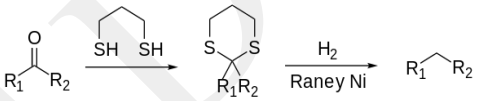
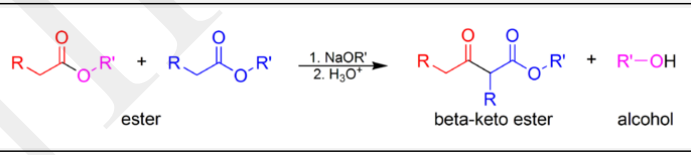
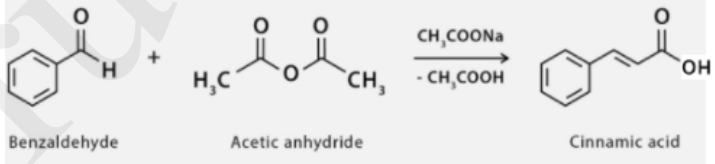
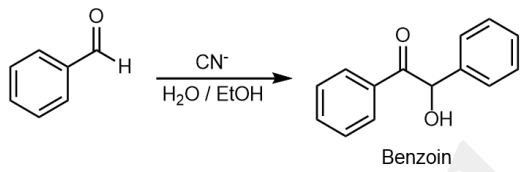
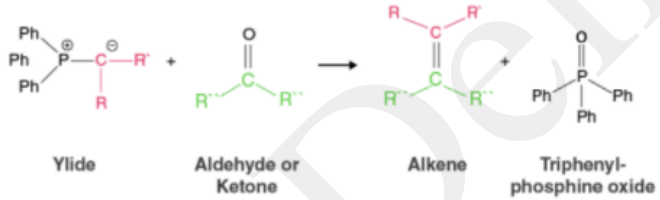
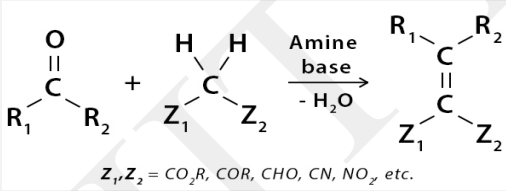
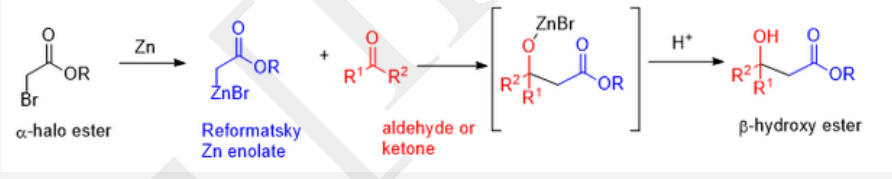
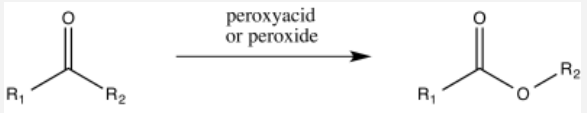
NAMED REACTIONS & REAGENTS IN ORGANIC CHEMISTRY

S. No	Named Reaction	Reagent	Use	Remark
1.	**Friedel Craft Alkylation**	RX + AlCl3	Alkyl group on benzene	No rkn on deactivated ring or with Ph-NH2, NHR, NR2 C+ rearrangement Order - RF > RCl > RBr > RI
2.	**Friedel Craft Acylation**	RCOCl + AlCl3	Acyl group on benzene	CO + HCl for -CHO No rearrangements
3.	**Clemmenson Reduction**	Zn-Hg conc HCl	C=O to CH2	Can reduce NO2
4.	**Wolff Kishner Reduction**	NH2NH2 in KOH	C=O to CH2	Cannot reduce NO2
5.	**Mozingo Reaction**	(i) SH-CH2CH2-SH in BF3 (ii) H2 Raney Ni	C=O to CH2 	Useful when comp is both acid and base sensitive
6.	**Corey House Synthesis**	Gillman Reagent - R2CuLi	R2CuLi + R'X -----> RR'	
7.	**Grignard**	RMgX	As base or Nu	To prepare CH2=CHMg***X*** or Ph-Mg***Cl*** solvent must be THF not DE
8.	**Birch reduction**	Na in Liq NH3	Trans addition of H2 across alkyne	
9.	**H2 + Lindlar's Catalyst**	Lindlar - {Pd/BaSO4 + Quinoline}	Syn addition of H2 across alkyne	
10.	**H2 + P2 Catalyst**	P2 Cat - Ni2B	Syn addition of H2 across alkyne	
11.	**Wurtz Reaction**	Na in DE	RX to R-R	CH3CH2X gives mixt of CH2=CH2 + CH3CH3 + CH3CH2CH2CH3
12.	**Fittig Reaction**	Na in DE	PhX to Ph-Ph	
13.	**Wurtz Fitting Reaction**	Na in DE	RX + Ph-X -----> R-Ph	
14.	**Kolbe's Electrolysis**	RCOONa + H2O	RCOONa gives R-R + CO2 + H2	CH3CH2X gives mixt of CH2=CH2 + CH3CH3 + CH3CH2CH2CH3
15.	**Bayer's Reagent**	Cold Alk KMnO4	Syn addition of OH across C=C	Pink/ Purple colour gets decolourised
16.	**FinkelStein Reaction**	NaI + Acetone	R-Cl/Br --------> R-I	NaBr/Cl are insoluble in acetone
17.	**Swartz Rkn**	AgF/ SbF3/ CoF2/ Hg2F2	RX --------> R-F	Aaj Sab ko hugg (Ag Sb Co Hg)

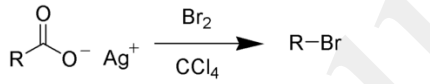
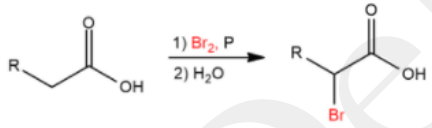
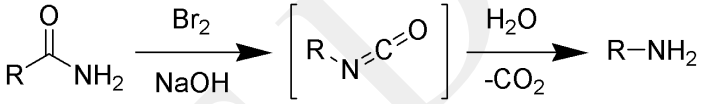
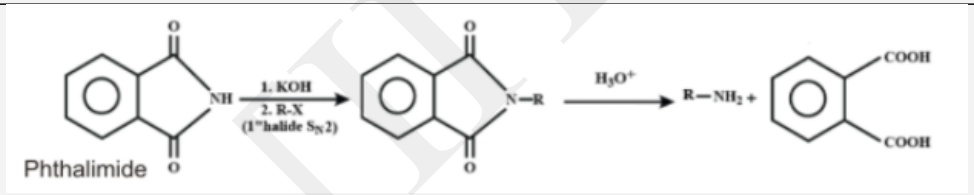
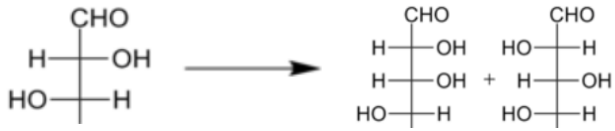
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18.	**Darzen Procedure**	SOCl2 in pyridine	ROH to RCl without rearrangement	Inversion of configuration
19.	**Reimer Tieman Reaction**	NaOH + CHCl3	Phenol to Salicadehyde	dichloro carbene is E+
20.	**Kolbe Schmidt Reaction**	NaOH +CO2	Phenol to Salicylicacid	
21.	**Williamson Synthesis of Ethers**	RL + RONa	Ethers prep	Elimination may take place
22.	**Rosenmund Reduction**	H2 + Lindlar's Catalyst	RCOCl to RCHO	
23.	**Etard's Reaction**	CrO2Cl2	Toulene to Ph-CHO	A bulkyl intermidate is formed as ppt preventing further [O]
24.	**Gatterman Koch Ald Synthesis**	CO + HCl + AlCl3 + CuCl	Benzene to Benzaldehyde	Not applicable to phenol (and their etheres); Deactivated rings; Benzene with NH2/ NHR/ NR2
25.	**Gatterman Aldehyde Reaction**	HCN + HCl + AlCl3	Benzene to Benzaldehyde	Applicable to phenol (and their etheres); Not applicable Deactivated rings; Benzene with NH2/ NHR/ NR2
26.	**Gatterman Formylation Reaction**	Zn(CN)2 + HCl	Benzene to Benzaldehyde	
27.	**Sommelet reaction (ADV+)**	(CH2)6N4	Ph-CH2Cl ----------> Ph-CHO	
28.	**Stephen's Reduction**	(i) SnCl2 + HCl (ii) H2O	RCN ---------> RCHO	
29.	**Aldol Condensation**	Dil NaOH	Ald/Ketones -----> β hydroxy Ald/Ketones	Must have α H; Can dehydrate easily; LDA will pick most acidic H (100% enolate formation no self aldol) while NaOH will form stable enol
30.	**Claisen Condensation**	RONa		2 α H are required. If only 1, then stronger bases like NaH,KH,NaNH2,Ph3CNa are required
31.	**Dieckmann Condesation**	(Intermolecular Claisen)		
32.	**Perkin Condensation**			Aliphatic Ald do not give this rkn (They have α H)

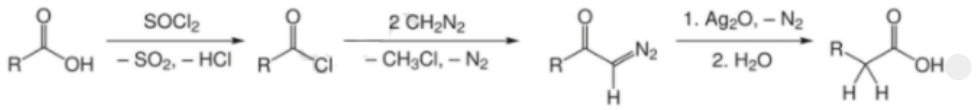
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33.	**Benzoin Condensation**	Alc KCN	 Benzoin	Nu attack on second CHO-Ph is RDS so, -CN will attack on the more activated ring first.
34.	**Cannizaro Reaction**	conc NaOH	R-CHO -------> RCOO- + RCH2OH	Disproportionation rkn; there shouldn't be any α H. RDS is transfer of hydride ion.
35.	**Wittig Rkn**	Ph3P=CH2		Stereoselective - trans > cis Regioselective - better than E2 (Double bond in just one place)
36.	**Tischchenko Reaction**	Al(OR)3	RCHO -------> RCOOR	
37.	**Knoevenagel Condensation**			
38.	**Reformatsky Reaction**			It is like Grignard, but much weaker so that it does not react with ester
39.	**Meerwein Ponndorf Verley (MPV) Reduction**	Al[OCH(CH3)2]3 + CH3CH(OH)CH3	Redeuce Ald/Ketones to Alcohols (especially in presence of NO2, C=C etc.)	Reverse rkn is called Opeenauer Oxidation
40.	**Baeyer Villiger Oxidation**	peracid like mcpBA		{ H > 3 > 2 >= Ph > 1 > CH3 } Migratory Aptitude for the group that will get attached to O (i.e. R2 has a greater MA than R1)

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41.	**Diel's Alder Reaction**			
42.	**Hunsdiecker Reaction**	Br2 + CCl4 + Heat		Yield of RX : 1 > 2 > 3 If RCOOAg and I2 are taken as 2:1 then ester is formed and in 1:1 R-I is formed **(Simonini Reaction)**
43.	**Hell Volhard Zelinsky reaction**	Red P + X2 + Heat		
44.	**Hoffman Bromamide Degradation (Rearrangement)**	NaOH + Br2		Intermediates : RCONH- RCONHBr -----> [RCONBr]- -------> R-N=C=O (Ketime) N loses both its H no Cross only self Only 1degree amines
45.	**Lossen Rearrangement**	(Same as Hoffman, in place of Br- we have -OCOR)		
46.	**Curtius Rearrangement**	(Same as Hoffman, in place of Br- we have -N2+)		
47.	**Gabriel Pthalimide Reaction**			Only 1 degree amines No Aromatic amines
48.	**Sandmeyer**	CuCl + HCl CuBr + HBr CuCN	On Benzene Diazonium salt to get halo/cyano benzene	
49.	**Gatterman**	Cu + HCl Cu + HBr	On Benzene Diazonium salt to get halobenzene	
50.	**Balz Schieman Reaction**	HBF4 + Heat	Benzene Diazonium to Fluorobenzene	
51.	**Kiliani Fischer Synthesis**	(i) HCN (ii) H3O+ (iii) Na-Hg (pH 3 to 5)		

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52.	**Arndt Eistert Reaction (Includes Wolff Rearrangement**		{O = C = CH - R} ---- Ketene is formed

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